CCSU BCA | SEMESTER I

Mathematical Foundation for Computer Science - I

Sets, relations, counting, graphs and matrices

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SYLLABUS	CCSU BCA 2025-26 / NEP

This paper combines discrete mathematics with matrix algebra. The strongest preparation is a mix of definitions, short proofs and fully shown numerical working.

Targeting 80+ is a study goal, not a guaranteed result. Use these notes with class material, the current syllabus and timed paper practice.

How to use this guide

Learn the short unit summary first. Then answer each question without looking, compare your answer with the model, and rewrite it in your own words. For calculations and programs, write every step and test with a small example.

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Unit I | Sets, relations and functions

Sets describe collections; relations describe links between elements; functions assign exactly one output to each input. The unit moves from set operations to relation properties, closures and function types.

Core ideas to remember

- For sets A and B, A x B contains ordered pairs (a,b). The order matters.
- A relation R on A is a subset of A x A. Check reflexive, symmetric and transitive properties separately.
- An equivalence relation is reflexive, symmetric and transitive; its equivalence classes partition the set.
- A function is injective if distinct inputs have distinct outputs, surjective if every codomain value is reached, and bijective if both.
- Warshall's algorithm builds reachability: for each intermediate k, set M[i,j] to 1 when M[i,k] and M[k,j] are both 1.

Important questions with model answers

Question: What makes a relation an equivalence relation? Give an example.

Model answer

A relation must be reflexive, symmetric and transitive. On the integers, define a R b when a-b is divisible by 3. Every integer is related to itself; if a-b is divisible by 3, so is b-a; and if a-b and b-c are divisible by 3, then a-c is divisible by 3. The classes are numbers with the same remainder on division by 3: {0,3,6,...}, {1,4,7,...}, and {2,5,8,...}, extending to negative integers.

Question: Explain Warshall's algorithm and what it produces.

Model answer

Start with the adjacency matrix of a directed graph. For each vertex k, treat k as a possible intermediate point. For every pair i,j, update M[i,j] = M[i,j] OR (M[i,k] AND M[k,j]). After all k values, M[i,j] is 1 exactly when a path from i to j exists. State whether the diagonal starts with zeros or ones according to whether the question asks for positive-length paths or reflexive reachability.

Unit II | Counting and recurrence relations

Counting uses the multiplication and addition principles, permutations, combinations, the pigeonhole principle and the binomial theorem. Recurrences describe sequences by earlier terms.

Core ideas to remember

- Use permutations when order matters: nPr = n!/(n-r)!; use combinations when order does not: nCr = n!/(r!(n-r)!).
- For distinct letters in COMPUTER, selecting and arranging four gives 8P4 = 8 x 7 x 6 x 5 = 1680.
- The pigeonhole principle says that placing more objects than boxes forces at least one box to hold two or more objects.
- For a linear recurrence with constant coefficients, form the characteristic equation and use its roots to construct the sequence solution.
- The binomial theorem is (x+y)^n = sum from r=0 to n of nCr x^(n-r)y^r.

Important questions with model answers

Question: How many distinct four-letter arrangements can be made from COMPUTER?

Model answer

The eight letters in COMPUTER are all different. Since order matters, use a permutation: 8P4 = 8!/(8-4)! = 8 x 7 x 6 x 5 = 1680. If a question allows repeated letters, or includes duplicate letters in the source word, adjust the counting method before calculating.

Question: Solve a simple recurrence: a(n)=3a(n-1)-2a(n-2), with a(0)=1 and a(1)=3.

Model answer

Assume a(n)=r^n. The characteristic equation is r^2-3r+2=0, so (r-1)(r-2)=0 and the roots are 1 and 2. Thus a(n)=A+B(2^n). From a(0)=1, A+B=1; from a(1)=3, A+2B=3. Therefore B=2 and A=-1, giving a(n)=2^(n+1)-1. Check: a(0)=1 and a(1)=3.

Unit III | Elementary graph theory

A graph models objects and the connections between them. Questions commonly test definitions, classification and small proofs or diagrams.

Core ideas to remember

- A graph has vertices and edges. A path follows adjacent vertices; a cycle returns to its start without repeating intermediate vertices.
- A tree is connected and acyclic. A tree with n vertices has n-1 edges.
- A spanning tree connects every vertex of a connected graph using the fewest possible edges and contains no cycle.
- An Euler circuit uses every edge once and returns to its starting vertex; a Hamiltonian cycle visits every vertex once and returns.
- For a connected planar graph, V-E+F=2, where F counts regions including the outside face.

Important questions with model answers

Question: Distinguish Euler and Hamiltonian circuits.

Model answer

An Euler circuit uses every edge exactly once and returns to its start; vertices may be visited more than once. A Hamiltonian cycle visits each vertex exactly once before returning to its start; it does not require using every edge. The distinction is whether the condition concerns edges or vertices. For a connected undirected graph, an Euler circuit exists when every vertex has even degree.

Question: What is a spanning tree? Explain why it has n-1 edges.

Model answer

A spanning tree is a connected, cycle-free subgraph containing all vertices of a connected graph. A basic tree with one vertex has zero edges. Each time a new vertex is added while preserving connectivity and avoiding cycles, exactly one edge is added. Therefore a tree with n vertices has n-1 edges. A spanning tree is useful for connecting network sites with no redundant links.

Unit IV | Matrix algebra

Matrix questions reward careful arithmetic and clear intermediate steps. Be fluent with determinants, rank, inverse, eigenvalues, Cramer's rule and the Cayley-Hamilton theorem.

Core ideas to remember

- A matrix inverse exists only when the determinant is non-zero.
- For a 2x2 matrix [[a,b],[c,d]], det = ad-bc and inverse = 1/(ad-bc) [[d,-b],[-c,a]].
- The rank is the number of linearly independent rows or columns; row reduction is a reliable way to find it.
- Eigenvalues satisfy det(A-lambda I)=0; an eigenvector satisfies (A-lambda I)v=0.
- Cramer's rule solves a square system when the determinant of its coefficient matrix is non-zero.

Important questions with model answers

Question: Find the eigenvalues and eigenvectors of [[2,1],[1,2]].

Model answer

Compute det(A-lambda I) = (2-lambda)^2-1 = 0. This gives lambda=3 or lambda=1. For lambda=3, the equations reduce to x=y, so one eigenvector is (1,1). For lambda=1, they reduce to x=-y, so one eigenvector is (1,-1). Any non-zero scalar multiple of either vector is also an eigenvector.

Question: Solve x+y=5 and 2x-y=1 using Cramer's rule.

Model answer

The coefficient determinant is D = |1 1; 2 -1| = -3. D_x = |5 1; 1 -1| = -6, so x=D_x/D=2. D_y = |1 5; 2 1| = -9, so y=D_y/D=3. Check both original equations: 2+3=5 and 4-3=1.

Paper pattern and topic analysis

The December 2025 paper transcription reports a 3-hour, 75-mark theory exam: five very short questions worth 3 marks each; attempt two of three 7.5-mark questions; and attempt three of five 15-mark questions. The CCSU syllabus lists 25 internal plus 75 external marks for this course.

Only one paper for the current 2025-26 course is listed, so a multi-year recurrence rate cannot be calculated. The 2025 paper sampled all four units: floor and ceiling, permutations and binomial expansion; Warshall and recurrence definitions; planar and spanning trees plus graph terminology; and inverse, rank, eigenvalues and Cramer's rule. That is evidence of broad unit coverage in that paper, not a guarantee for the next sitting.

What to prioritize

- Practice Warshall updates, relation properties and function classifications.
- Solve permutations, combinations, pigeonhole and binomial questions without skipping the setup.
- Draw and label graphs; compare Euler and Hamiltonian properties and use Euler's planar formula.
- Show full determinant, inverse, rank, Cramer's rule and eigenvalue working.

Last-five-year archive check

The CCSUStudy BCA archive lists the current Mathematical Foundation for Computer Science-I paper for December 2025 (archive label 19007). For 2021-2024, it lists Mathematics-I papers under the earlier BCA syllabus, including 2021 variants and 2022, 2023 and 2024 papers. Those older papers are not treated as the current course because their calculus-heavy syllabus differs. Use an old question only where its topic is explicitly present in this syllabus, especially matrices.

How to read the evidence

The 2025 question themes and section pattern are drawn from a third-party paper transcription and cross-checked against the archive listing. Unit importance and model answers are based on the published 2025-26 syllabus. Frequency claims across multiple years are not made.

Learning resources and source list

The paper archive is a discovery aid, not the university's official syllabus. The current course titles and units below follow the 2025-26 CCSU syllabus reproduced by an affiliated college. Older paper titles are identified as legacy or related material wherever they differ.

CCSU BCA syllabus and unit list (2025-26 reproduction)

CCSUStudy BCA question-paper archive

December 2025 Mathematical Foundation paper listing

NPTEL Discrete Mathematics for CS, IIT Kanpur

NPTEL Graph Theory, IISER Pune

Coverage note: no question bank can predict the exact paper. Treat the priority list as a revision map and prepare every listed unit.